Organisation of Cells

Exam code: 0610 & 0970

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Producing New Cells

- The cells in your body need to be able to divide to help your body **grow** and **repair** itself
- Cells grow and divide over and over again
- New cells are produced by the **division of existing cells**

Did this video help you? Yes No

Specialised Cells

Specialised cells in animals

- Specialised cells are those which have **developed certain characteristics** in order to **perform particular functions**. These differences are controlled by genes in the nucleus
- Cells specialise by undergoing **differentiation**: this is a process by which cells develop the structure and characteristics needed to be able to carry out their functions

Specialised Cells in Animals Table

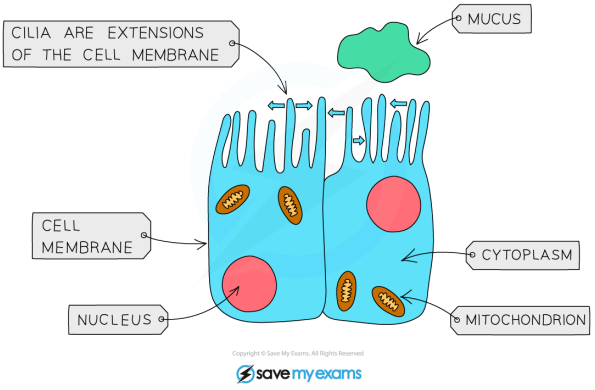
Stuck on something?

Ask questions and get help with exactly what you're revising.

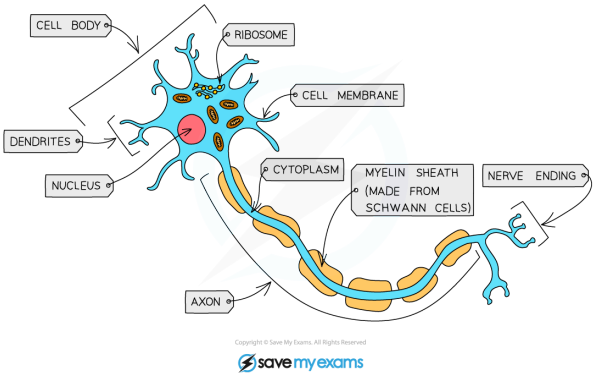
Got it

	TRACHEA AND BRONCHI	**CILIA** WHICH **BEAT TO MOVE MUCUS** AND TRAPPED PARTICLES UP TO THE THROAT
NERVE CELL	CONDUCTION OF IMPULSES	– **LONG** SO THAT NERVES CAN RUN TO AND FROM **DIFFERENT PARTS OF THE BODY** TO THE CENTRAL NERVOUS SYSTEM – THE CELL HAS **EXTENSIONS AND BRANCHES**, SO THAT IT CAN **COMMUNICATE** WITH OTHER NERVE CELLS, MUSCLES AND GLANDS – THE AXON (EXTENSION OF CYTOPLASM AWAY FROM THE CELL BODY) IS COVERED WITH A **FATTY SHEATH**, WHICH **INSULATES** THE NERVE CELL AND **SPEEDS UP** THE NERVE IMPULSE
RED BLOOD CELL	TRANSPORT OF OXYGEN	– **BICONCAVE DISC** SHAPE **INCREASES SURFACE AREA** FOR MORE EFFICIENT DIFFUSION OF OXYGEN – CONTAINS **HAEMOGLOBIN** WHICH **JOINS WITH OXYGEN** TO TRANSPORT IT – CONTAINS **NO NUCLEUS** TO **INCREASE AMOUNT OF SPACE** AVAILABLE FOR HAEMOGLOBIN INSIDE CELL
SPERM CELL	REPRODUCTION	– THE HEAD CONTAINS THE GENETIC MATERIAL FOR FERTILISATION IN A **HAPLOID NUCLEUS** (CONTAINING HALF THE NORMAL NUMBER OF CHROMOSOMES) – THE **ACROSOME** IN THE HEAD CONTAINS **DIGESTIVE ENZYMES** SO THAT A SPERM CAN PENETRATE AN EGG – THE MID-PIECE IS PACKED WITH **MITOCHONDRIA** TO RELEASE **ENERGY** NEEDED TO SWIM AND FERTILISE THE EGG – THE **TAIL** ENABLES THE SPERM TO **SWIM**
EGG CELL (OVUM)	REPRODUCTION	– CONTAINS A **LOT OF CYTOPLASM** WHICH HAS **NUTRIENTS** FOR THE GROWTH OF THE EARLY EMBRYO – **HAPLOID NUCLEUS** CONTAINS THE GENETIC MATERIAL FOR FERTILISATION – CELL MEMBRANE **CHANGES AFTER FERTILISATION** BY A SINGLE SPERM SO THAT **NO MORE SPERM** CAN ENTER

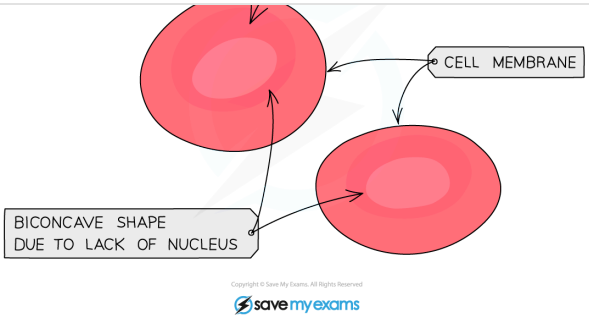
Diagrams of specialised cells in animals:



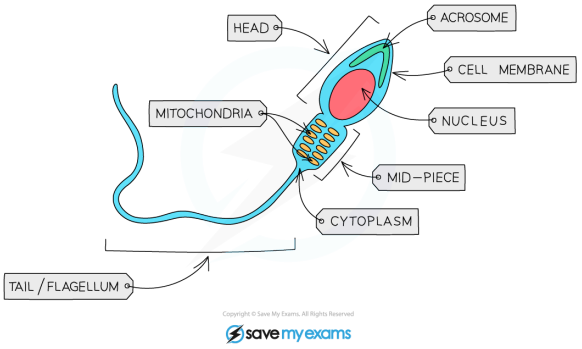
Ciliated cell



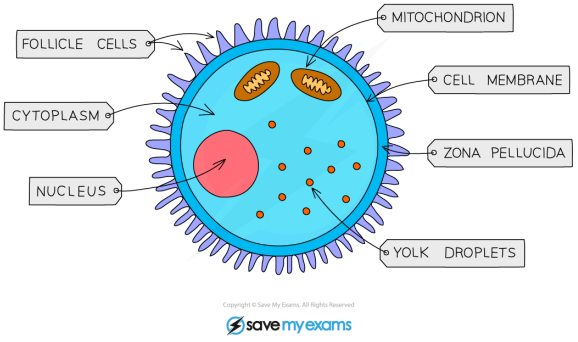
Nerve cell



Red blood cells



Sperm cell

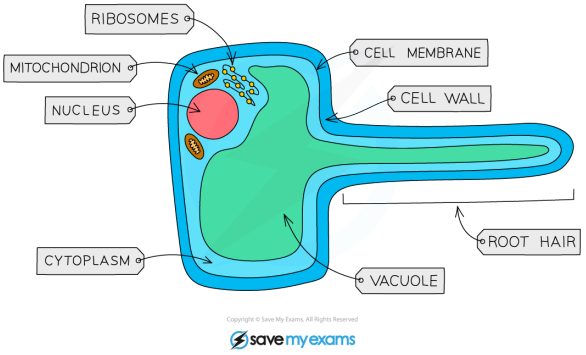


Egg cell

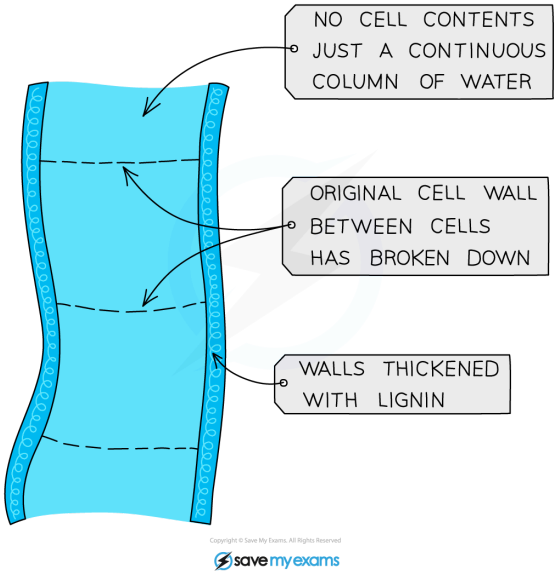
Examples of specialised cells in plants:

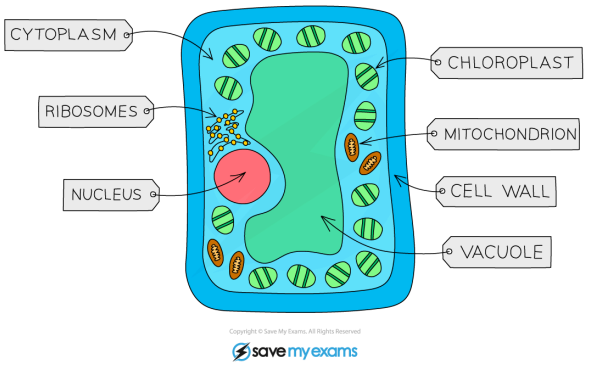
	FROM SOIL	WATER AND MINERAL IONS – WALLS ARE **THIN** TO ENSURE WATER MOVES THROUGH **QUICKLY** – NO CHLOROPLASTS PRESENT
XYLEM VESSEL	CONDUCTION OF WATER THROUGH THE PLANT; SUPPORT OF THE PLANT	– **NO TOP AND BOTTOM WALLS** BETWEEN XYLEM VESSELS, SO THERE IS A **CONTINUOUS COLUMN OF WATER** RUNNING THROUGH THEM – CELLS ARE **DEAD** WITHOUT ORGANELLES OR CYTOPLASM TO ALLOW **FREE PASSAGE OF WATER** – THEIR WALLS BECOME **THICKENED** WITH A SUBSTANCE CALLED **LIGNIN** WHICH MEANS THEY ARE ABLE TO HELP **SUPPORT** THE PLANT
PALISADE MESOPHYLL CELL	PHOTOSYNTHESIS	– **COLUMN SHAPED** TO MAXIMIZE ABSORPTION OF SUNLIGHT AND **FIT AS MANY IN A LAYER** UNDER THE UPPER EPIDERMIS OF THE LEAF AS POSSIBLE – CONTAINS **MANY CHLOROPLASTS** FOR MAXIMUM **PHOTOSYNTHESIS**

Diagrams of specialised cells in plants:



Root hair cell





Palisade mesophyll cell

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Did this video help you? Yes No

Levels of Organisation in an Organism

Levels of organisation

- Your syllabus states that you should be able to identify the different levels of organisation in drawings, diagrams and images of familiar material
- An example of this is shown in the exam question below:

The answer is B

Examiner Tips and Tricks

Most incorrect answers here come from not being able to identify a tissue, so it's worth making sure you understand and remember that **tissues are always made up of only one type of cell**.

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